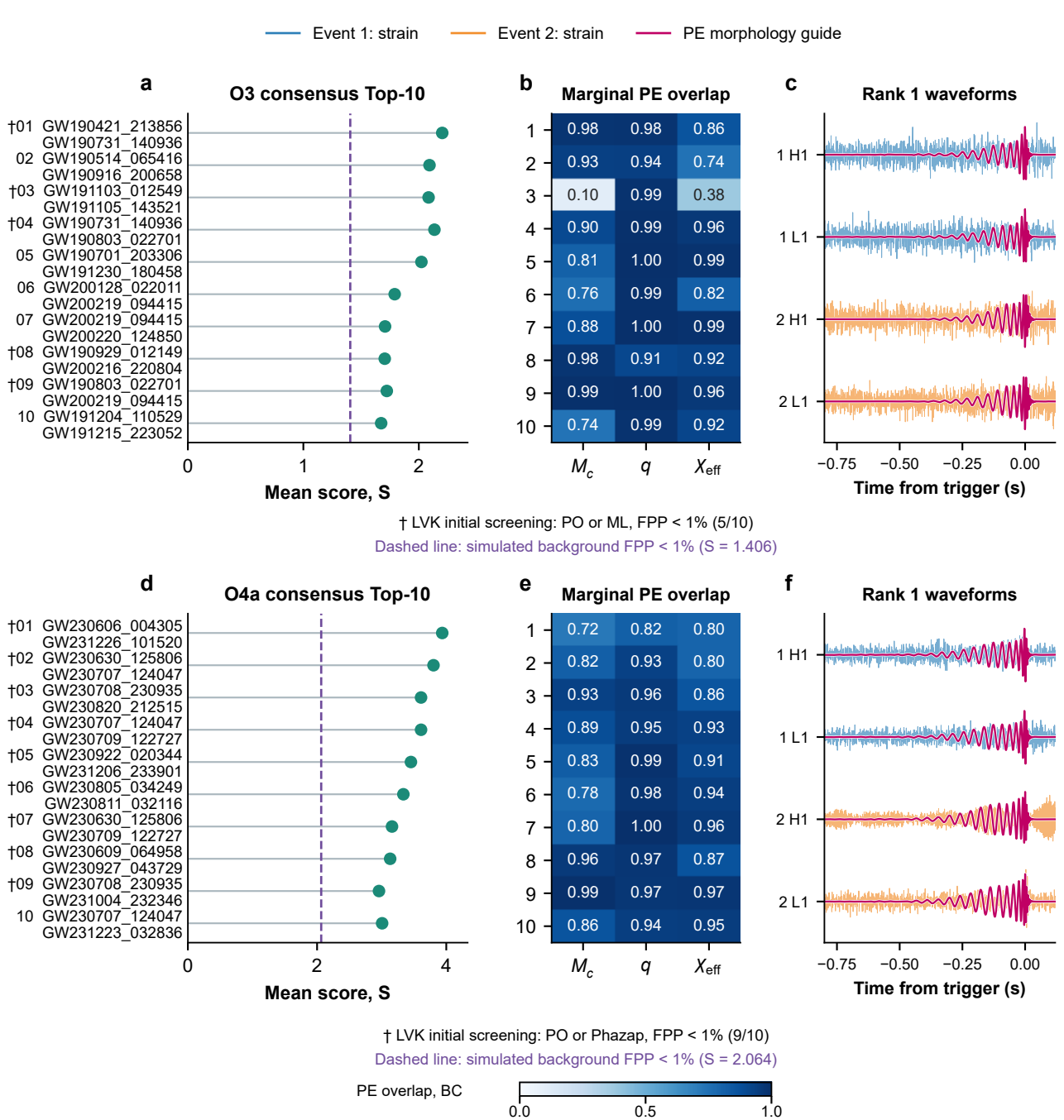

Event 1: strain
Event 2: strain
PE morphology guide
a
O3 consensus Top-10
†01 GW190421_213856
GW190731_140936
02 GW190514_065416
GW190916_200658
†03 GW191103_012549
GW191105_143521
†04 GW190731_140936
GW190803_022701
05 GW190701_203306
GW191230_180458
06 GW200128_022011
GW200219_094415
07 GW200219_094415
GW200220_124850
†08 GW190929_012149
GW200216_220804
†09 GW190803_022701
GW200219_094415
10 GW191204_110529
GW191215_223052
0
1
2
Mean score, S
b
Marginal PE overlap
1 0.98 0.98 0.86
2 0.93 0.94 0.74
3 0.10 0.99 0.38
4 0.90 0.99 0.96
5 0.81 1.00 0.99
6 0.76 0.99 0.82
7 0.88 1.00 0.99
8 0.98 0.91 0.92
9 0.99 1.00 0.96
10 0.74 0.99 0.92
M_c
q
X_{eff}
c
Rank 1 waveforms
1 H1
1 L1
2 H1
2 L1
−0.75
−0.50
−0.25
0.00
Time from trigger (s)
† LVK initial screening: PO or ML, FPP < 1% (5/10)
Dashed line: simulated background FPP < 1% (S = 1.406)
d
O4a consensus Top-10
†01 GW230606_004305
GW231226_101520
†02 GW230630_125806
GW230707_124047
†03 GW230708_230935
GW230820_212515
†04 GW230707_124047
GW230709_122727
†05 GW230922_020344
GW231206_233901
†06 GW230805_034249
GW230811_032116
†07 GW230630_125806
GW230709_122727
†08 GW230609_064958
GW230927_043729
†09 GW230708_230935
GW231004_232346
10 GW230707_124047
GW231223_032836
0
2
4
Mean score, S
e
Marginal PE overlap
1 0.72 0.82 0.80
2 0.82 0.93 0.80
3 0.93 0.96 0.86
4 0.89 0.95 0.93
5 0.83 0.99 0.91
6 0.78 0.98 0.94
7 0.80 1.00 0.96
8 0.96 0.97 0.87
9 0.99 0.97 0.97
10 0.86 0.94 0.95
M_c
q
X_{eff}
f
Rank 1 waveforms
1 H1
1 L1
2 H1
2 L1
−0.75
−0.50
−0.25
0.00
Time from trigger (s)
† LVK initial screening: PO or Phazap, FPP < 1% (9/10)
Dashed line: simulated background FPP < 1% (S = 2.064)
PE overlap, BC
0.0
0.5
1.0